

Yukhta Seelam

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SUMMARY

I'm interested in developing **autonomous robotic systems** that can perceive, reason, and make **reliable decisions** in dynamic environments. My work has focused on **reinforcement learning** for autonomous driving, **robot perception**, and learning-based control, with experience building and evaluating intelligent systems in simulation using **ROS**, **CARLA**, and deep learning. I'm particularly interested in bridging machine learning with robotics to build autonomous systems that are robust, efficient, and **deployable in the real world**.

ACADEMIC PROFILE

Boston University Sep 2024 - Jan 2026
Masters in Robotics and Autonomous Systems

Indian Institute of Technology Kanpur Sep 2020 - May 2024
B. Tech in Electrical Engineering

SKILLS & COURSEWORK

- **Programming:** Python, C++, MATLAB, Bash, SQL
- **ML/DL/AI Frameworks:** LangChain, Pydantic, PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost
- **Robotics Tools:** ROS, Gazebo, RViz, CARLA, OpenCV, SLAM, Control Systems, Path Planning
- **Hardware & Design:** Fusion 360, SolidWorks, Eagle, Arduino, Embedded Systems
- **Specializations:** Deep Learning, LLMs, Reinforcement Learning, Federated Learning, Generative Models, Computer Vision

EXPERIENCE & RESEARCH

AI and ML Engineer Jan 2026 – Present
Holiday Channel - Boston, MA (Remote)

- Designed and deployed a **RAG-based AI chatbot** architecture integrating web data ingestion, text chunking, embedding generation, **vector search** (ChromaDB), and **LLM-based** response generation for grounded customer support.
- Built automated **ETL workflows** to scrape and preprocess holiday knowledge from websites using **BeautifulSoup** and **Playwright**, transforming unstructured HTML into **searchable document embeddings**.
- Developed a **Streamlit**-based chatbot prototype leveraging **semantic search** and LLM inference to deliver context-aware responses while minimizing model hallucinations through **retrieval-grounded generation**.

Autonomous Driving Researcher Feb 2025 – Sep 2025
Red Hat - Boston, MA

- Developed and trained a **hybrid local-cloud reinforcement** learning policy in the **CARLA simulator**, optimizing **inference latency by 30 %** and reducing **compute energy by 22 %** in autonomous driving scenarios.
- Resolved headless **rendering** issues and implemented a video logging pipeline by using **OpenCV** backends, enabling visualization of training and evaluation rollouts.
- Conducted large-scale training with **checkpoint recovery**, implemented custom progress logging, and performed **policy evaluation** using RGB, depth, and segmentation sensor data.

Robotics Engineer May 2024 – Sep 2024
Samsung Research & Development Institute - Delhi, India

- Developed **CNN-based perception models** (YOLOv5, RefineDet) for robotic vacuum cleaners, improving **object detection** accuracy for real-world autonomous navigation.
- Designed and integrated obstacle detection modules for identifying household objects such as cables, furniture, and floor-level clutter to **enhance collision avoidance** and **navigation reliability**.
- Collaborated with robotics and ML teams to optimize and deploy **vision models** across multiple hardware platforms, ensuring reliable performance in diverse indoor environments.

Robotics Research Intern Aug 2023 – Mar 2024
Indian Institute of Technology Kanpur - Kanpur, India

- Developed perception-based obstacle avoidance algorithms for the rover by integrating **real-time sensor data** with **ROS-based A* and TEB planners**, enabling safe **navigation** through dynamic and unstructured environments.
- Implemented a **YOLOv5-based** armor plate **classification** system, achieving **85% mAP**, and integrated it with a **PID-controlled gimbal** for accurate target tracking and autonomous shooting while the rover was in motion.

AWARDS AND PUBLICATIONS

- 3rd Place – IEEE ICRA DJI RoboMaster AI Challenge 2022
- Paper: "Towards Autonomous Shooting Rover via Situation-Aware Visual Perception and Dynamic Action Execution", IEEE/SICE SII 2024 (Vietnam)
- Poster: "Design of AI Ground Robots for Intelligent Combat in a Dynamic Arena", ICRA 2022